



An innate quinone functionalized electrochemically exfoliated graphene/Fe₃O₄ composite electrode for the continuous generation of reactive oxygen species

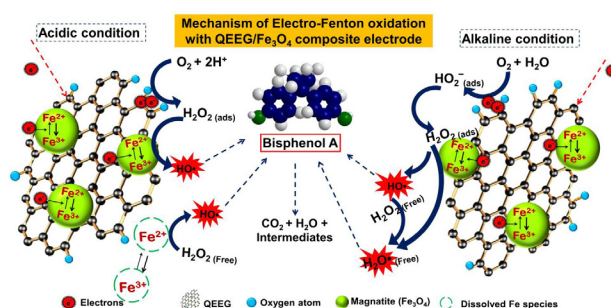
Govindaraj Divyapriya, Indumathi M. Nambi^{*}, Jaganathan Senthilnathan

Environmental and Water Resources Division, Department of Civil Engineering, Indian Institute of Technology Madras, Chennai 600036, India

HIGHLIGHTS

- Single step/direct synthesis of quinone functionalized graphene from graphite.
- Substantial improvement of H₂O₂ production was observed with QEEG electrode.
- Optimized loading of Fe₃O₄ on QEEG electrode showed better removal of Bisphenol A.
- Stable catalytic activity was observed for >10 cycles of reuse.
- Cleaner and greener process; recommended for the treatment of drinking water.

GRAPHICAL ABSTRACT



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ABSTRACT

This study successfully demonstrated the controlled synthesis of innate quinone-functionalized graphene by an eco-friendly electrochemical exfoliation process. A quinone-functionalized electrochemically exfoliated graphene (QEEG) and QEEG/Fe₃O₄ electrode was established for the continuous electro-generation of H₂O₂ and ·OH radicals. The quality of the exfoliated graphene was assessed by Raman spectroscopy. The nature of functional groups and the oxidation states of QEEG and QEEG/Fe₃O₄ composite were evaluated with X-ray photoelectron spectroscopy (XPS). High-resolution Transmission Electron Microscopic (HR-TEM) analysis shows that the Fe₃O₄ nanoparticles (<10 nm) are uniformly dispersed on the surface of QEEG. The presence of a quinone functional group on QEEG domain and its role in oxygen reduction reaction for the formation of H₂O₂ in acidic, neutral and alkaline conditions were examined. Significant enhancement in the production of H₂O₂ was observed with QEEG electrode. Optimized QEEG/Fe₃O₄ electrode achieved complete removal of Bisphenol A within 90 min at pH 3.0. Degradation efficiency of 98% was observed at neutral pH with less than 1% of iron leaching. Stable catalytic activity was observed even after 10 cycles of the experiments. On the basis of experimental results, the possible electro-Fenton oxidation mechanism with QEEG/Fe₃O₄ composite cathode was proposed. Cleaner water treatment process was successfully demonstrated with no chemical addition and negligible iron contamination.

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1. Introduction

In recent decades, the occurrence and persistence of emerging contaminants in aquatic system have turned out to be the key

^{*} Corresponding author.

E-mail address: indunambi@iitm.ac.in (I.M. Nambi).